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JAXtronomy: A JAX port of lenstronomy

Alan Huang [1], **Simon Birrer** [1], **Natalie B. Hogg** [2], **Aymeric Galan** [3,4], **Daniel Gilman** [5], **Anowar J. Shajib** [5,6,7], **and Nan Zhang** [8]

1 Department of Physics and Astronomy, Stony Brook University, Stony Brook, NY 1794, USA **2** Laboratoire Univers et Particules de Montpellier, CNRS and Université de Montpellier (UMR-5299), 34095 Montpellier, France **3** Max-Planck-Institut für Astrophysik, Karl-Schwarzschild Straße 1, 85748 Garching, Germany **4** Technical University of Munich, TUM School of Natural Sciences, Physics Department, James-Franck-Straße 1, 85748 Garching, Germany **5** Department of Astronomy and Astrophysics, University of Chicago, Chicago, Illinois 60637, USA **6** Kavli Institute for Cosmological Physics, University of Chicago, Chicago, IL 60637, USA **7** Center for Astronomy, Space Science and Astrophysics, Independent University, Bangladesh, Dhaka 1229, Bangladesh **8** Department of Physics, University of Illinois, 1110 West Green St., Urbana, IL 61801, USA

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Summary

Gravitational lensing is a phenomenon where light bends around massive objects, resulting in distorted images seen by an observer. Studying gravitationally lensed systems provides insights into cosmology and astrophysics, including constraints of the expansion rate of the Universe and the distribution of dark matter.

Thus, we introduce `JAXtronomy`, a re-implementation of the gravitational lensing software package `lenstronomy` (Birrer, 2021; Birrer & Amara, 2018) using `JAX` (Bradbury et al., 2018). `JAX` is a Python library that uses an accelerated linear algebra (XLA) compiler to improve the performance of computing software. Our core design principle of `JAXtronomy` is to maintain an identical API to that of `lenstronomy`.

The main `JAX` features utilized in `JAXtronomy` are just-in-time-compilation, which can lead to significant reductions in execution time, and automatic differentiation, which allows for the implementation of gradient-based algorithms that were previously impossible. Additionally, `JAX` allows code to be run on GPUs or parallelized across CPU cores, further boosting the performance of `JAXtronomy`.

Statement of need

`lenstronomy` has been widely applied to numerous science cases, with more than 200 publications making use of the software, and has an increasing number of dependent packages relying on features of `lenstronomy`. For instance, science cases directly involving `lenstronomy` include galaxy evolution studies using strong lensing (Shajib et al., 2021; Sheu et al., 2025; Tan et al., 2024) and detailed lens modeling for measuring the Hubble constant using time-delay cosmography by the TDCOSMO collaboration (Birrer et al., 2020, 2025; Birrer & Treu, 2021; Gilman et al., 2020; Millon et al., 2020; Schmidt et al., 2025; Shajib et al., 2022; Williams et al., 2025).

Examples of packages dependent on `lenstronomy` for general-purpose lensing computations and image modelling include the `dolphin` package (Shajib et al., 2025) for automated lens modeling, the `galight` package (Ding et al., 2020) for galaxy morphology measurements, `SLSim` (Khadka et al, 2026, in prep) for simulating large populations of strong lenses, `pyHalo` (Gilman et al., 2019) and `mejiro` (Wedig et al., 2025) for simulating strong lenses with dark

matter substructure, and PALTAS (Wagner-Carena et al., 2023) for neural network inference tasks.

In many of these applications, computational constraints are the key limiting factor for strong gravitational lensing science. For example, increased data quality and number of lenses to analyze makes lens modeling a computational bottleneck, and expensive ray-tracing through tens of thousands of dark matter substructures limit the amount of images that can be simulated, especially for the training of neural networks and simulation-based inferences. These ever-increasing computational costs have lead to the development of several JAX-accelerated and GPU-accelerated strong-lensing packages, such as `gigalens` (Gu et al., 2022), `herculens` (Vernardos et al., 2022), `paltax` (Wagner-Carena et al., 2024), `GLaD` (Wang et al., 2025), `caustics` (Stone et al., 2024) and Google Research's `jaxstronomy` (Google Research, 2023).

Why JAXtronomy?

`JAXtronomy` inherits a wide range of features from `lenstronomy` that are not offered by any of the aforementioned JAX-accelerated or GPU-accelerated software. These features include `lenstronomy`'s linear amplitude solver, which reduces the number of sampled parameters during lens modeling, as well as a variety of log likelihood functions and optional punishment terms to improve robustness during fitting. `JAXtronomy` aims to maintain an identical API to `lenstronomy` so that packages dependent on `lenstronomy` can transition seamlessly to `JAXtronomy`.

Improvements over lenstronomy in image simulation

The simulation of a lensed image comes in three main steps. The first step begins with a coordinate grid in the angles seen by the observer. These coordinates are ray-traced through the deflectors back to the source plane. This process requires the calculation of light ray deflection angles at each deflector. Second, the surface brightness of the source is calculated on the ray-traced coordinate grid. This produces a lensed image. Third, the lensed image gets convolved by the point spread function (PSF) originating from diffraction of the telescope optics and atmospheric turbulence. Due to the various choices in deflector mass profiles, light model profiles, grid size, and PSF kernel size, the overall runtime of the pipeline can vary significantly.

In the following sections, we outline the improvements in performance that `JAXtronomy` has over `lenstronomy` for each step in the pipeline. These performance benchmarks were run using an Intel(R) Xeon(R) Gold 6338 CPU @ 2.00GHz, an NVIDIA A100 GPU, and `JAX` version 0.7.0.

Deflection angle calculations

Each entry in the table indicates how much faster `JAXtronomy` is compared to `lenstronomy` at computing deflection angles for the corresponding deflector profile and grid size. Some comparisons vary significantly with values of function arguments, so a range is given rather than a number.

Deflector Profile	60x60 grid (cpu)	180x180 grid (cpu)	180x180 grid (gpu)
CONVERGENCE	0.4x	1.1x	0.5x
CSE	1.6x	2.6x	2.6x
EPL	5.1x - 15x	9.2x - 17x	37x - 120x
EPL (jax) vs EPL_NUMBA	1.4x	3.0x	13x
EPL_MULTIPOLE_M1M3M4	2.1x - 7x	6.4x - 13x	42x - 108x
HERNQUIST	2.0x	3.4x	5.8x

Deflector Profile	60x60 grid (cpu)	180x180 grid (cpu)	180x180 grid (gpu)
HERNQUIST_ELLIPSE_CSE	3.8x	5.4x	40x
MULTIPOLE	0.9x	1.0x	8.3x - 14x
MULTIPOLE_ELL	1.5x - 2.1x	2.0x - 2.8x	70x
NIE/SIE	0.5x	0.5x	2.0x
NFW	1.6x	3.3x	4.5x
NFW_ELLIPSE_CSE	4.1x	6.7x	31x
PJAFFE	1.0x	1.2x	2.8x
PJAFFE_ELLIPSE_POTENTIAL	1.4x	1.6x	3.1x
SHEAR	0.7x	2.0x	0.9x
SIS	1.4x	3.3x	2.0x
TNFW	2.4x	5.8x	7.5x

Due to `JAX`'s higher function call overheads compared to standard Python, slowdowns can occur when using computationally simple deflector profiles. These deflector profiles benefit most from the parallelization that `JAX` offers.

Flux calculations

An analogous table for the different light profiles is shown below. The MULTI_GAUSSIAN and MULTI_GAUSSIAN_ELLIPSE profiles include five GAUSSIAN and GAUSSIAN_ELLIPSE components, respectively, highlighting JAX's improved performance in sequential computations.

Light Profile	60x60 grid (cpu)	180x180 grid (cpu)	180x180 grid (gpu)
CORE_SERSIC	2.0x	6.7x	4.2x
GAUSSIAN	1.0x	2.5x	1.3x
GAUSSIAN_ELLIPSE	1.5x	3.6x	2.0x
MULTI_GAUSSIAN	3.7x	11x	7.8x
MULTI_GAUSSIAN_ELLIPSE	4.0x	13x	6.9x
SERSIC	1.0x	1.7x	3.9x
SERSIC_ELLIPSE	1.9x	5.7x	3.2x
SERSIC_ELLIPSE_Q_PHI	1.7x	5.5x	3.3x
SHAPELETS (n_max=6)	6.2x	3.4x	15x
SHAPELETS (n_max=10)	6.0x	4.5x	17x

FFT Convolution

We find that FFT convolution using `JAX` on CPU results in variable performance boosts or slowdowns compared to `lenstronomy` (which uses `scipy`'s FFT convolution). On a 60x60 grid, and kernel sizes ranging from 3 to 45, `JAX` on CPU ranges from being 1.1x to 2.9x faster than `lenstronomy`, with no obvious correlation to kernel size. On a 180x180 grid, and kernel sizes ranging from 9 to 135, `JAXtronomy` on CPU ranges from being 0.7x to 2.5x as fast as `lenstronomy`, with no obvious correlation to kernel size.

However, FFT convolution using `JAX` on GPU is significantly faster than `scipy`. On a 60x60 grid, and kernel sizes ranging from 3 to 45, JAX on GPU ranges from being 1.5x to 3.5x faster than `lenstronomy`, with `JAX` performing better at higher kernel sizes. On a 180x180 grid, and kernel sizes ranging from 9 to 135, `JAXtronomy` on GPU is about 10x to 20x as fast as `lenstronomy`, again with `JAX` performing better at higher kernel sizes.

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Improvements over lenstronomy in lens modelling

The process of lens modelling involves finding best-fit parameters describing a lensed system from real data. In `lenstronomy`, this typically involves a Particle Swarm Optimizer (PSO) (Kennedy & Eberhart, 1995) for optimization and Monte Carlo Markov Chains for posterior sampling. `JAXtronomy` retains these lens modelling algorithms from `lenstronomy` while benefitting from the increased performance outlined above.

In the following table, we compare `JAXtronomy`'s PSO performance to that of `lenstronomy` when modeling a lens with a singular isothermal ellipsoid (SIE) mass profile, Sersic-ellipse source and lens light profile, and a quadruply-imaged point source. The image is simulated using a 100x100 grid and FFT convolved using a PSF kernel with a size of 13 pixels.

Device	64 Particles	128 Particles	256 Particles	512 Particles
lenstronomy (baseline)	59s	138s	245s	555s
1 CPU core	3x	3x	3x	4x
2 CPU cores	5x	6x	6x	7x
4 CPU cores	8x	12x	11x	12x
8 CPU cores	11x	17x	17x	24x
16 CPU cores	13x	20x	22x	29x
32 CPU cores	13x	20x	20x	29x
GPU	8x	7x	27x	46x

Additionally, using `JAX`'s autodifferentiation, we have implemented the L-BFGS gradient descent algorithm from the `Optax` library (Babuschkin et al., 2020) for optimization. This is a significant improvement over `lenstronomy`'s PSO, which does not have access to gradient information. Due to the stochastic nature of the PSO, we do not present a concrete comparison between `lenstronomy`'s PSO and `JAXtronomy`'s minimizer for the time it takes to find best-fit parameters.

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References

Babuschkin, I., Baumli, K., Bell, A., Bhupatiraju, S., Bruce, J., Buchlovsky, P., Budden, D., Cai, T., Clark, A., Danihelka, I., Dedieu, A., Fantacci, C., Godwin, J., Jones, C., Hemsley, R., Hennigan, T., Hessel, M., Hou, S., Kapturowski, S., … Viola, F. (2020). *The DeepMind JAX Ecosystem*. http://github.com/google-deepmind

Bingham, E., Chen, J. P., Jankowiak, M., Obermeyer, F., Pradhan, N., Karaletsos, T., Singh, R., Szerlip, P. A., Horsfall, P., & Goodman, N. D. (2019). Pyro: Deep universal probabilistic programming. *J. Mach. Learn. Res.*, *20*, 28:1–28:6. http://jmlr.org/papers/v20/18-403.html

Birrer, S. (2021). Gravitational lensing formalism in a curved arc basis: A continuous description of observables and degeneracies from the weak to the strong lensing regime. *The Astrophysical Journal*, *919*(1), 38. https://doi.org/10.3847/1538-4357/ac1108

DRAFT

Birrer, S., & Amara, A. (2018). Lenstronomy: Multi-purpose gravitational lens modelling software package. *Physics of the Dark Universe*, *22*, 189–201. https://doi.org/10.1016/j.dark.2018.11.002

Birrer, S., Buckley-Geer, E. J., Cappellari, M., Courbin, F., Dux, F., Fassnacht, C. D., Frieman, J. A., Galan, A., Gilman, D., Huang, X.-Y., Knabel, S., Langeroodi, D., Lin, H., Millon, M., Morishita, T., Motta, V., Mozumdar, P., Paic, E., Shajib, A. J., … Wong, K. C. (2025). TDCOSMO 2025: Cosmological constraints from strong lensing time delays. *Astronomy & Astrophysics*, *704*, A63. https://doi.org/10.1051/0004-6361/202555801

Birrer, S., Shajib, A. J., Galan, A., Millon, M., Treu, T., Agnello, A., Auger, M., Chen, G. C.-F., Christensen, L., Collett, T., Courbin, F., Fassnacht, C. D., Koopmans, L. V. E., Marshall, P. J., Park, J.-W., Rusu, C. E., Sluse, D., Spiniello, C., Suyu, S. H., … Van de Vyvere, L. (2020). TDCOSMO - IV. Hierarchical time-delay cosmography – joint inference of the hubble constant and galaxy density profiles⋆. *Astronomy & Astrophysics*, *643*, A165. https://doi.org/10.1051/0004-6361/202038861

Birrer, S., & Treu, T. (2021). TDCOSMO - v. Strategies for precise and accurate measurements of the hubble constant with strong lensing. *Astronomy & Astrophysics*, *649*, A61. https://doi.org/10.1051/0004-6361/202039179

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). *JAX: Composable transformations of Python+NumPy programs* (Version 0.3.13). http://github.com/jax-ml/jax

Ding, X., Silverman, J., Treu, T., Schulze, A., Schramm, M., Birrer, S., Park, D., Jahnke, K., Bennert, V. N., Kartaltepe, J. S., Koekemoer, A. M., Malkan, M. A., & Sanders, D. (2020). The Mass Relations between Supermassive Black Holes and Their Host Galaxies at $1 < z < 2$ HST-WFC3. *The Astrophysical Journal*, *888*(1), 37. https://doi.org/10.3847/1538-4357/ab5b90

Gilman, D., Birrer, S., Nierenberg, A., Treu, T., Du, X., & Benson, A. (2019). Warm dark matter chills out: Constraints on the halo mass function and the free-streaming length of dark matter with eight quadruple-image strong gravitational lenses. *Monthly Notices of the Royal Astronomical Society*, *491*(4), 6077–6101. https://doi.org/10.1093/mnras/stz3480

Gilman, D., Birrer, S., & Treu, T. (2020). TDCOSMO - III. Dark matter substructure meets dark energy. The effects of (sub)halos on strong-lensing measurements of H0. *Astronomy & Astrophysics*, *642*, A194. https://doi.org/10.1051/0004-6361/202038829

Google Research. (2023). *Jaxstronomy: JAX-based tools for astronomical inference*. https://github.com/google-research/google-research/tree/master/jaxstronomy

Gu, A., Huang, X., Sheu, W., Aldering, G., Bolton, A. S., Boone, K., Dey, A., Filipp, A., Jullo, E., Perlmutter, S., Rubin, D., Schlafly, E. F., Schlegel, D. J., Shu, Y., & Suyu, S. H. (2022). GIGA-lens: Fast bayesian inference for strong gravitational lens modeling. *The Astrophysical Journal*, *935*(1), 49. https://doi.org/10.3847/1538-4357/ac6de4

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Kennedy, J., & Eberhart, R. (1995). Particle swarm optimization. *Proceedings of ICNN'95 - International Conference on Neural Networks*, *4*, 1942–1948 vol.4. https://doi.org/10.1109/ICNN.1995.488968

Millon, M., Galan, A., Courbin, F., Treu, T., Suyu, S. H., Ding, X., Birrer, S., Chen, G. C.-F., Shajib, A. J., Sluse, D., Wong, K. C., Agnello, A., Auger, M. W., Buckley-Geer, E. J.,

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Chan, J. H. H., Collett, T., Fassnacht, C. D., Hilbert, S., Koopmans, L. V. E., … Van de Vyvere, L. (2020). TDCOSMO - i. An exploration of systematic uncertainties in the inference of H0 from time-delay cosmography. *Astronomy & Astrophysics*, *639*, A101. https://doi.org/10.1051/0004-6361/201937351

Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in NumPyro. *arXiv Preprint arXiv:1912.11554*.

Schmidt, T., Treu, T., Birrer, S., Millon, M., Sluse, D., Galan, A., Shajib, A. J., Lemon, C., Dux, F., & Courbin, F. (2025). TDCOSMO. XVIII. Strong lens model and time-delay predictions for J1721+8842, the first einstein zigzag lens. *Astronomy & Astrophysics*. https://doi.org/10.1051/0004-6361/202449984

Shajib, A. J., Nihal, N. S., Tan, C. Y., Sahu, V., Birrer, S., Treu, T., & Frieman, J. (2025). Dolphin: A fully automated forward-modeling pipeline powered by artificial intelligence for galaxy-scale strong lenses. *The Astrophysical Journal*, *992*(1), 40. https://doi.org/10.3847/1538-4357/adf95c

Shajib, A. J., Treu, T., Birrer, S., & Sonnenfeld, A. (2021). Dark matter haloes of massive elliptical galaxies at z $\sim$ 0.2 are well described by the Navarro-Frenk-White profile. *Monthly Notices of the Royal Astronomical Society*, *503*(2), 2380–2405. https://doi.org/10.1093/mnras/stab536

Shajib, A. J., Wong, K. C., Birrer, S., Suyu, S. H., Treu, T., Buckley-Geer, E. J., Lin, H., Rusu, C. E., Poh, J., Palmese, A., Agnello, A., Auger-Williams, M. W., Galan, A., Schuldt, S., Sluse, D., Courbin, F., Frieman, J., & Millon, M. (2022). TDCOSMO. IX. Systematic comparison between lens modelling software programs: Time-delay prediction for WGD 2038−4008. *Astronomy & Astrophysics*, *667*, A123. https://doi.org/10.1051/0004-6361/202243401

Sheu, W., Shajib, A. J., Treu, T., Sonnenfeld, A., Birrer, S., Cappellari, M., Oldham, L. J., & Tan, C. Y. (2025). Project Dinos II: redshift evolution of dark and luminous matter density profiles in strong-lensing elliptical galaxies across $0.1 < z < 0.9$. *Monthly Notices of the Royal Astronomical Society*, *541*(1), 1–27. https://doi.org/10.1093/mnras/staf976

Stone, C., Adam, A., Coogan, A., Yantovski-Barth, M. J., Filipp, A., Setiawan, L., Core, C., Legin, R., Wilson, C., Barco, G. M., Hezaveh, Y., & Perreault-Levasseur, L. (2024). Caustics: A python package for accelerated strong gravitational lensing simulations. *Journal of Open Source Software*, *9*(103), 7081. https://doi.org/10.21105/joss.07081

Tan, C. Y., Shajib, A. J., Birrer, S., Sonnenfeld, A., Treu, T., Wells, P., Williams, D. M., Buckley-Geer, E. J., Drlica-Wagner, A., & Frieman, J. (2024). Project Dinos I: A joint lensing-dynamics constraint on the deviation from the power law in the mass profile of massive ellipticals. *Monthly Notices of the Royal Astronomical Society*, *530*(2), 1474–1505. https://doi.org/10.1093/mnras/stae884

Vernardos, G., Peel, A., Courbin, F., & Starck, J.-L. (2022). Using wavelets to capture deviations from smoothness in galaxy-scale strong lenses. *Astronomy & Astrophysics*, *668*, A155. https://doi.org/10.1051/0004-6361/202244464

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., … SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, *17*, 261–272. https://doi.org/10.1038/s41592-019-0686-2

Wagner-Carena, S., Aalbers, J., Birrer, S., Nadler, E. O., Darragh-Ford, E., Marshall, P. J., & Wechsler, R. H. (2023). From images to dark matter: End-to-end inference of substructure from hundreds of strong gravitational lenses. *The Astrophysical Journal*, *942*(2), 75. https://doi.org/10.3847/1538-4357/aca525

Wagner-Carena, S., Lee, J., Pennington, J., Aalbers, J., Birrer, S., & Wechsler, R. H.

DRAFT

(2024). A strong gravitational lens is worth a thousand dark matter halos: Inference on small-scale structure using sequential methods. *The Astrophysical Journal*, *975*(2), 297. https://doi.org/10.3847/1538-4357/ad6e70

Wang, H., Suyu, S. H., Galan, A., Halkola, A., Cappellari, M., Shajib, A. J., & Cernetic, M. (2025). GPU-accelerated gravitational lensing and dynamical (GLaD) modeling for cosmology and galaxies. *Astronomy & Astrophysics*, *701*, A280. https://doi.org/10.1051/0004-6361/202554861

Wedig, B., Daylan, T., Birrer, S., Cyr-Racine, F.-Y., Dvorkin, C., Finkbeiner, D. P., Huang, A., Huang, X., Karthik, R., Khadka, N., Natarajan, P., Nierenberg, A. M., Peter, A. H. G., Pierel, J. D. R., Tang, X. T., & Wechsler, R. H. (2025). The roman view of strong gravitational lenses. *The Astrophysical Journal*, *986*(1), 42. https://doi.org/10.3847/1538-4357/adc24f

Williams, D. M., Treu, T., Birrer, S., Shajib, A. J., Wong, K. C., Morishita, T., Schmidt, T., & Stiavelli, M. (2025). TDCOSMO - XX. WFI2033–4723, the first quadruply imaged quasar modeled with JWST imaging. *Astronomy & Astrophysics*, *703*, A118. https://doi.org/10.1051/0004-6361/202554359

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